

VPN Guide

Paolo Joseph Baioni

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1 Introduction

The Mathematics Department VPN portal is gp-dmat-saml.vpn.polimi.it. VPN access is required to use HPC resources. The procedure varies depending on the user's operating system (OS); details follow.

Remark The access request for the student cluster implicitly includes access to the VPN if the requester does not already have it. In particular, BSc and MSc students are added to a specific authorisation group that allows access only to their HPC login node. Consequently, traffic to some resources may be restricted while connected to the VPN.

Remark VPN requests are not processed instantaneously by the online services.

2 Windows and macOS users

For Windows and macOS users, follow the official Polimi instructions at <https://www.ict.polimi.it/network/vpn/?lang=en>. For usage instructions, refer to section 3.3.

3 GNU/Linux users

At the time of writing, GlobalProtect (GP) officially supports only certain releases of Red Hat Enterprise Linux and Ubuntu; some combinations of OS version and GP version are known to cause issues, especially with Wi-Fi. However, unless you are using an end-of-life distribution, a Free and Open Source command-line solution is recommended for its simplicity and the positive results observed in testing on various distributions.

3.1 FOSS CLI solution

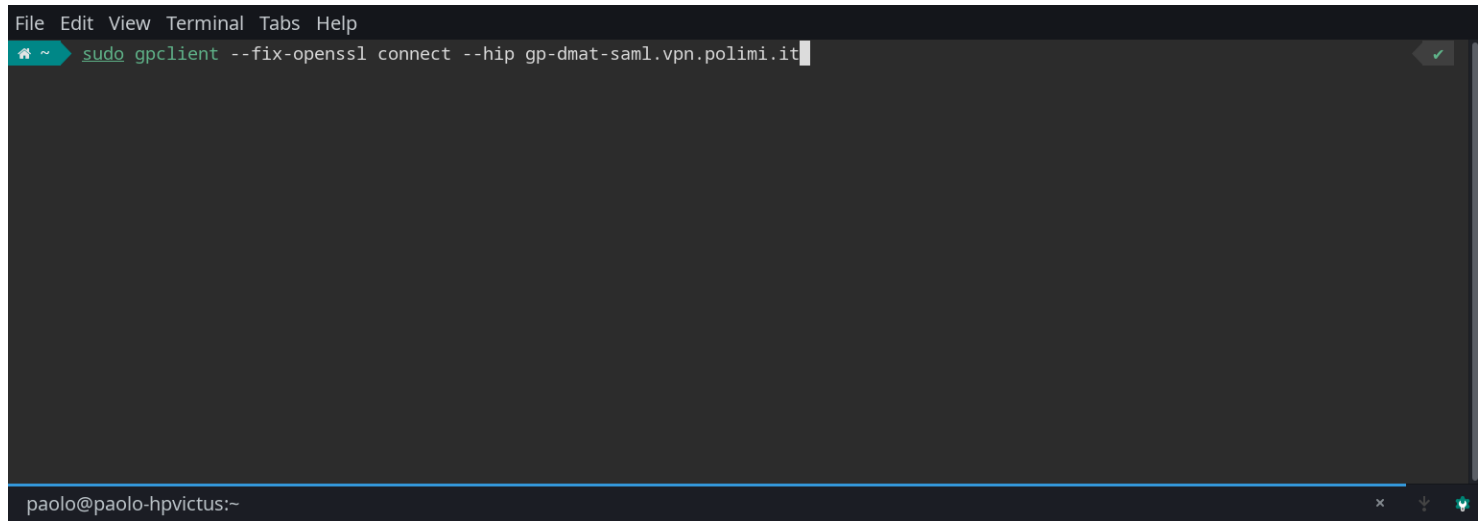
The software is available at: <https://github.com/yuezk/GlobalProtect-openconnect.git> which also contains detailed installation instructions.

Remark If your distribution is not officially listed in the README but uses a package system that is described there, you can often install it by downloading the appropriate archive and installing any missing dependencies when prompted. For example, on RHEL 10 you can install the RPM package via dnf, provided the packages that dnf requests are installed (also via dnf).

To start the connection, the base command is:

```
gpclient connect gp-dmat-saml.vpn.polimi.it
```

Some additional flags may be required, as shown in the figure:



```
File Edit View Terminal Tabs Help
~ sudo gpclient --fix-openssl connect --hip gp-dmat-saml.vpn.polimi.it
paolo@paolo-hpvictus:~
```

Figure 1:

The software will trigger Polimi two-factor authentication (2FA) twice; in both cases the usual procedure applies.



GlobalProtect Login

 **POLITECNICO MILANO 1863**

SP VPN Dipartimento Matematica

 **Entra con SPID**

 **Entra con CIE**

 **Login with eIDAS**

 **Entra con eduGAIN**

Credenziali Polimi

Accedi [Registrati](#)

☐ Resta connesso. ⓘ

▼ **Assistenza autenticazione**

[E' il tuo primo accesso?](#)

[Recupero credenziali Polimi \(codice persona e password\).](#)

[Disattiva 2FA: accedi temporaneamente con le sole Credenziali Polimi](#)

[Informazioni, domande frequenti e assistenza](#)

Figure 2:

Remark For both SPID and CIE, the QR code may be hidden by default.

GlobalProtect Login

MINISTERO DELL'INTERNO

CARTELLA DI IDENTITÀ ELETTRONICA

RICHIESTA D'ACCESSO CIE LIVELLO 2

Politecnico di Milano

Entra con le tue credenziali CIE

Numero CIE o Codice fiscale o Email ⓘ

Password ⓘ

Hai dimenticato la password? [Richiedine una nuova.](#)

[Annulla](#) [Procedi](#)

Figure 3:

If so, maximise the window to show it:

GlobalProtect Login

MINISTERO DELL'INTERNO

CARTELLA DI IDENTITÀ ELETTRONICA

RICHIESTA D'ACCESSO CIE LIVELLO 2

Politecnico di Milano

Entra con le tue credenziali CIE

Numero CIE o Codice fiscale o Email ⓘ

Password ⓘ

Hai dimenticato la password? [Richiedine una nuova.](#)

[Annulla](#) [Procedi](#)

Non hai ancora attivato le credenziali CIE? [Attivale ora](#)

Entra con App CielD

Se hai attivato le Credenziali CIE (livello 2) e hai certificato il dispositivo, apri l'App CielD ed inquadra il QR Code.

Se non vuoi usare la modalità semplificata ed hai già registrato la tua carta su App CielD (livello 3), seleziona "Entra con lettura carta CIE" qui sotto.

Validità codice: 1'36"

oppure

Entra con lettura carta

Puoi entrare con la lettura della Carta (livello 3) da uno smartphone NFC con l'App CielD registrando la tua carta o da PC con il software CIE ed un lettore di smart card.

[Scopri di più.](#)

[Entra con lettura carta](#)

Figure 4:

Once the connection is established, the terminal will show output similar to:

```
[2025-10-20T11:22:43Z INFO openconnect::ffi] Using vhost-net for tun acceleration, ring size 32
[2025-10-20T11:22:43Z INFO openconnect::vpn] Connected to VPN, pipe_fd: 12
[2025-10-20T11:22:43Z INFO gpclient::connect] Wrote PID 2213 to /var/run/gpclient.lock

sudo gpclient --fix-openssl connect --hip gp-dmat-saml.vpn.polimi.it
```

Figure 5:

This indicates that the VPN connection is active.

Remark At this point, open a new terminal window or tab to connect to the cluster via ssh.
To close the VPN connection, press Ctrl+D:

```
File Edit View Terminal Tabs Help
[2025-10-21T14:45:27Z INFO openconnect::ffi] GlobalProtect HIP check due
[2025-10-21T14:45:27Z INFO openconnect::ffi] POST https://gp-dmat-saml.vpn.polimi.it/ssl-vpn/hipreportcheck.esp
[2025-10-21T14:45:27Z INFO openconnect::ffi] SSL negotiation with gp-dmat-saml.vpn.polimi.it
[2025-10-21T14:45:27Z INFO openconnect::ffi] Connected to HTTPS on gp-dmat-saml.vpn.polimi.it with ciphersuite (TLS1.2)-(ECDHE-SECP256R1)
~(RSA-SHA256)-(AES-256-GCM)
^C[2025-10-21T15:01:46Z INFO gpclient::connect] Received the interrupt signal, disconnecting...
[2025-10-21T15:01:46Z INFO openconnect::ffi] Stopping VPN connection: 12
[2025-10-21T15:01:46Z INFO openconnect::ffi] POST https://gp-dmat-saml.vpn.polimi.it/ssl-vpn/logout.esp
[2025-10-21T15:01:46Z INFO openconnect::ffi] SSL negotiation with gp-dmat-saml.vpn.polimi.it
[2025-10-21T15:01:46Z INFO openconnect::ffi] Connected to HTTPS on gp-dmat-saml.vpn.polimi.it with ciphersuite (TLS1.2)-(ECDHE-SECP256R1)
~(RSA-SHA256)-(AES-256-GCM)
[2025-10-21T15:01:46Z INFO openconnect::ffi] Logout successful.
resolvconf: signature mismatch: /etc/resolv.conf
resolvconf: run 'resolvconf -u' to update
[2025-10-21T15:01:46Z INFO openconnect::ffi] openconnect_mainloop returned -4, exiting
[2025-10-21T15:01:46Z INFO gpclient::connect] Removing PID file
~ >
2h 17m 16s 17:01:46
paolo@paolo-hpvictus:~
```

Figure 6:

3.2 GlobalProtect_UI

3.2.1 Installation

If you are using an older but officially supported GP distribution, getting GlobalProtect-openconnect to work can be more challenging than installing it. Follow the official Palo Alto Networks instructions:

<https://docs.paloaltonetworks.com/globalprotect/5-1/globalprotect-app-user-guide/globalprotect-app-for-linux/download-and-install-the-globalprotect-app-for-linux>

For example, on RHEL 9.6 you can install the graphical client with:

```
sudo dnf install ./GlobalProtect_UI_rpm-6.0.7.1-372.rpm
```

which installs the graphical tool.

On Ubuntu 22.04 LTS:

```
sudo apt install --fix-missing ./GlobalProtect_deb-6.0.7.1-372.deb
```

It may also be useful to run:

```
systemctl --user daemon-reload
```

3.3 Usage

Open the app by clicking its icon:

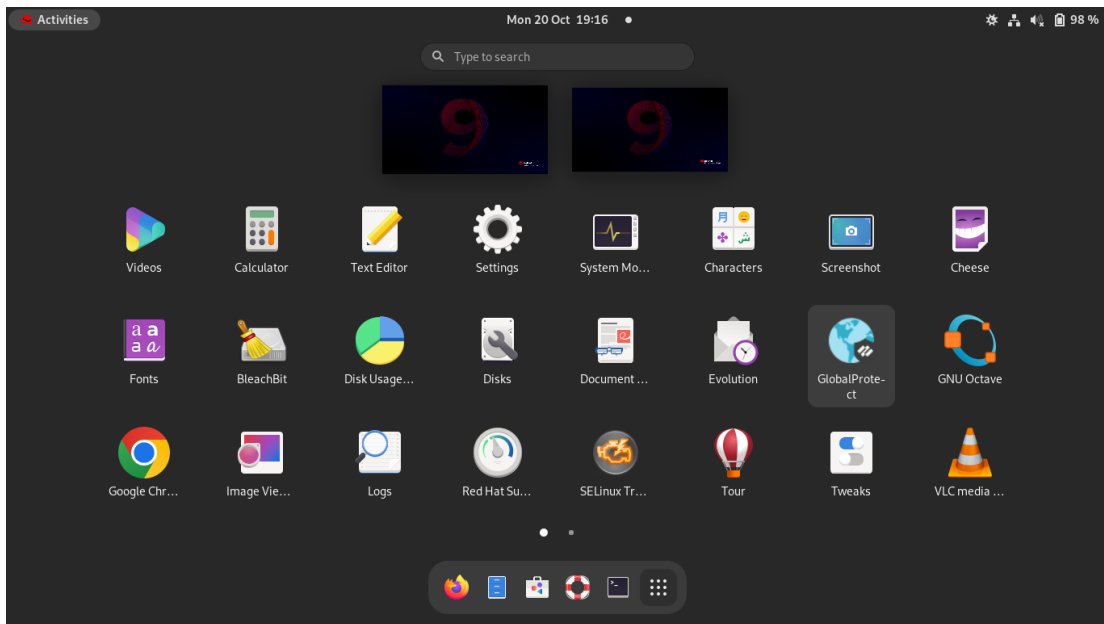


Figure 7:

The app will show as not connected; the first time you must configure the settings:

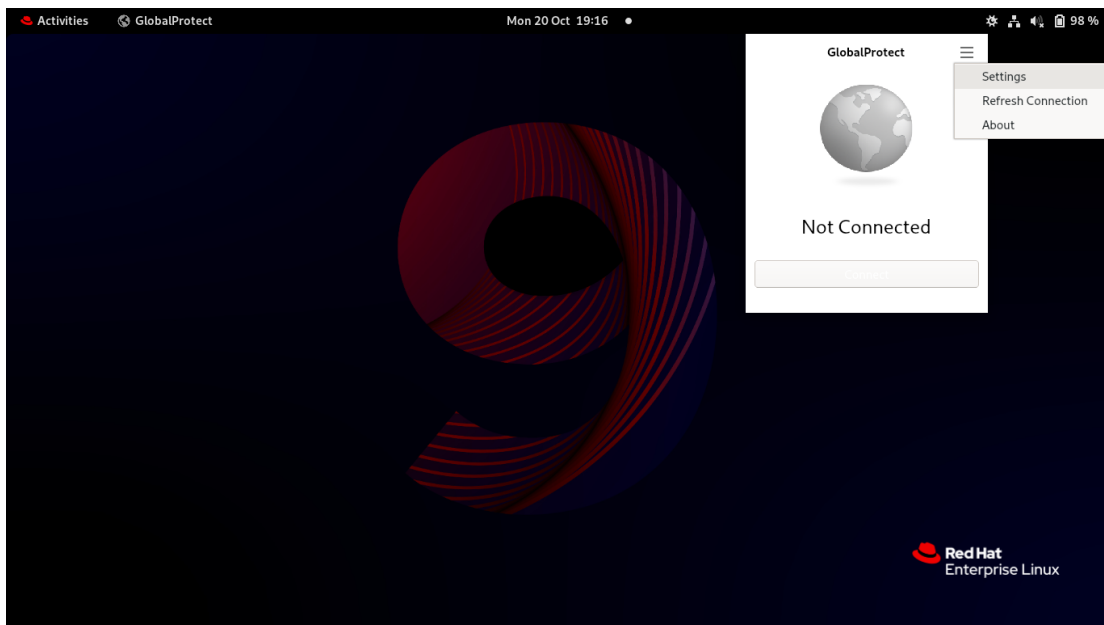


Figure 8:

Enter the portal as shown:

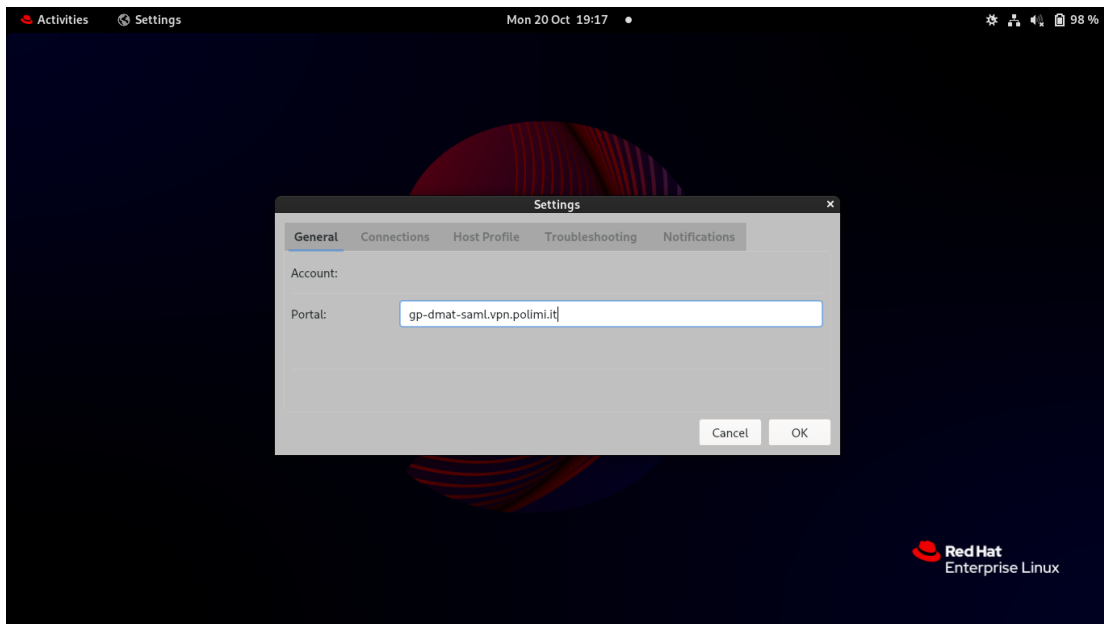


Figure 9:

Save and click Connect:

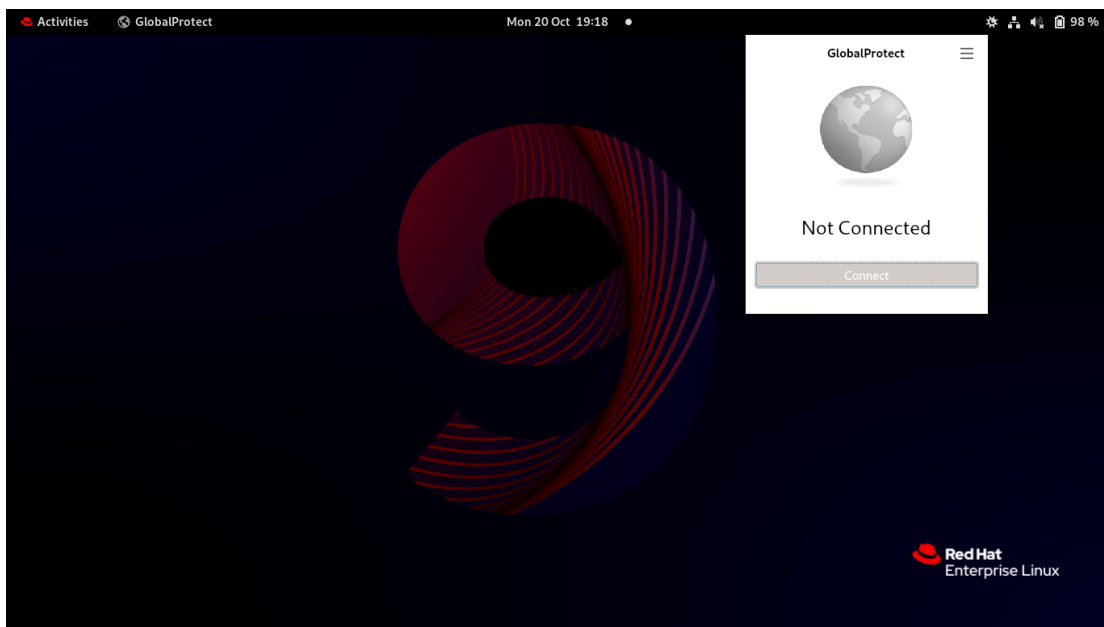


Figure 10:

This will open your default browser; click "Open GlobalProtect" when prompted:

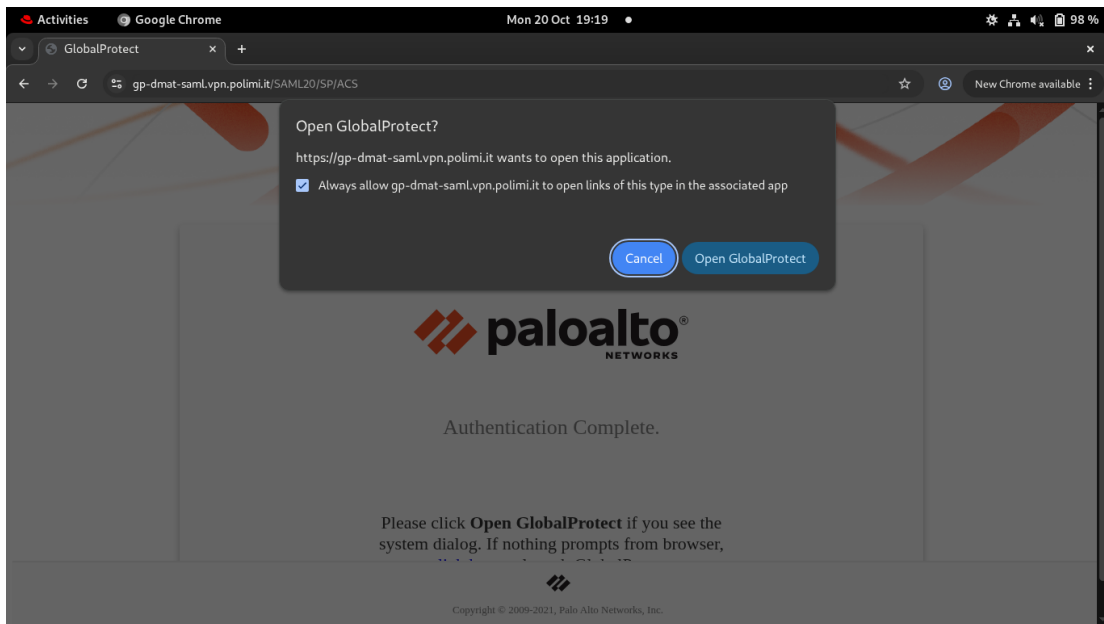


Figure 11:

Then click the provided link:

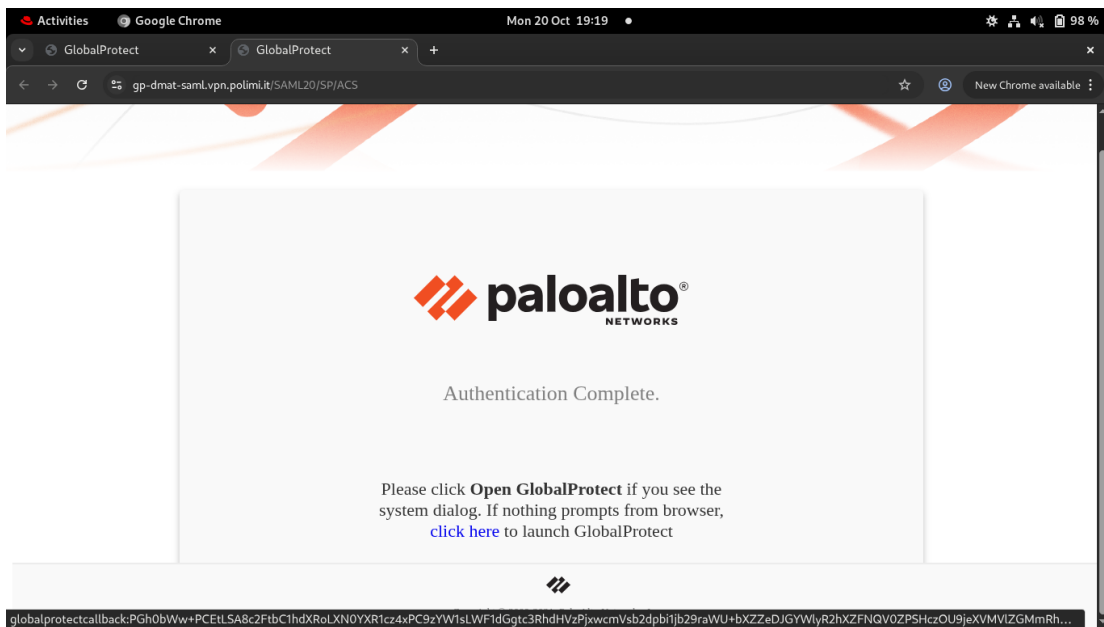


Figure 12:

You can verify a successful connection by reopening GlobalProtect:

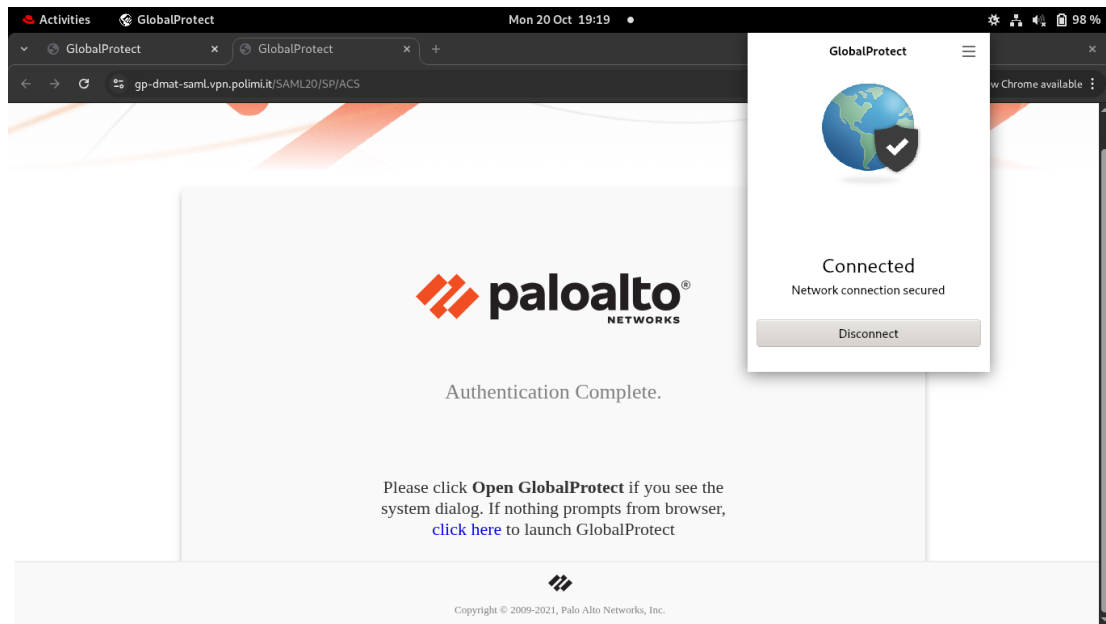


Figure 13: